

The DAC 2012 Routability-Driven Placement Contest and Benchmark Suite

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ABSTRACT

Existing routability-driven placers mostly employ rudimentary and often crude congestion models that fail to account for the complexities in modern designs, e.g., the impact of non-uniform wiring stacks, layer directives, partial and/or complete routing blockages, etc. In addition, they are hampered by congestion metrics that do not accurately score or represent design congestion. This is in large part due to the non-availability of public designs depicting industrial wiring stacks and other complexities affecting design routability.

The aim of the DAC 2012 routability-driven placement contest is to address these issues, by way of the following: (a) release challenging benchmark designs that are derived from modern industrial ASICs, and contain information to perform both placement and routing, (b) present a new congestion metric, as well as an accurate congestion analysis framework to evaluate and compare the routability of various placement algorithms. We hope that a set of challenging benchmarks, along with a standard, publicly available evaluation framework will further advance research in routability-driven placement.

Categories and Subject Descriptors

B.7.2 [Hardware, Integrated Circuits, Design Aids]: Placement and routing

General Terms

Algorithms, Design, Experimentation, Performance

Keywords

Physical Design, Placement, Routing, Benchmarks, Congestion Analysis

1. INTRODUCTION

Design routability has emerged as one of the primary concerns during physical synthesis at advanced technology

nodes [1]. In this regard, placement is perhaps one of the most powerful tools to help mitigate routing congestion during a physical synthesis flow. However, optimizing for traditional placement metrics like the total half-perimeter wire length (used in the ISPD-2005 contest [13]) or density-target based overflow (used in the ISPD-2006 contest [12]) are no longer sufficient to address this critical issue. Modern placers need to explicitly account for congestion to obtain routable placements.

Prior work (e.g., [2, 7, 16]) has incorporated some form of congestion modeling within global placement. The key issue with these techniques is that they employ rudimentary and often inaccurate congestion models like pin-density or probabilistic congestion estimation. These models fail to account for the complexities in modern designs.

To predict design congestion with reasonable accuracy, modern congestion analysis techniques need to account for:

- A complex wiring stack with anywhere from nine to twelve metal layers, along with varying width and spacing, within and across layers.
- Partial and/or complete routing blockages at various metal layers.
- Local wiring congestion due to *intra*-g-cell routes, and pin-access issues.
- Layer directives, e.g., from optimization tools.

Prior work (e.g., [10, 14, 15, 21]) has also tried to address the issue of congestion modeling by using reasonably accurate global routing based congestion analyzers – that in part handle the complex wiring constraints of modern designs. However, they are hampered by congestion metrics that often do not accurately score or represent the design congestion [18].

This paper describes the DAC 2012 routability-driven placement contest. It also presents a set of benchmarks that are representative of modern industrial designs with numerous placement and routing blockages, more metal layers, varying width and spacing across layers, etc. The primary objectives of the DAC 2012 routability-driven placement contest are:

- Release advanced benchmarks derived from modern industrial ASIC designs to advance research in the areas of placement and routing for nanometer-scale VLSI designs.
- Motivate research in the area of fast, yet reasonably accurate routing congestion analysis that can handle the complex features and constraints of modern wiring stacks.

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- Motivate research to effectively integrate placement with accurate congestion analysis techniques to perform routability-driven placement.
- Provide an accurate congestion analysis framework, by releasing a set of global routing based congestion analyzers that also model local wiring or local routing resource usage.
- Present intuitive and accurate metrics to score and compare the routability of various placement algorithms.

The rest of this paper is organized as follows: Section 2 gives an overview of the new congestion metric that is used for the DAC 2012 contest. Section 3 presents the DAC 2012 contest evaluation metric. Section 4 describes the DAC 2012 benchmark suite. Finally, Section 5 provides concluding remarks.

2. CONGESTION ANALYSIS

Congestion analysis refers to the process of estimating the routability of a given placement or design point. Typically, the output of this process is a congestion or hot-spot map indicating the regions in the chip with the greatest contention for wiring resources. In a physical synthesis flow, the effectiveness of various routability-driven optimizations (e.g., placement spreading, logic re-synthesis, etc.) are highly dependent on the speed and accuracy of congestion analysis.

2.1 Global Routing Based Congestion Analysis

With the advent of fast and high-quality global routers [5, 8, 9, 11, 19, 20], global routing based congestion analysis offers an attractive compromise between run-time and accuracy, and is gaining widespread usage. *Hence, global routing based congestion analysis was the method of choice to evaluate the routability of the placement solutions during the DAC 2012 contest.* In this regard, every design in the DAC 2012 benchmark suite (Sec. 4) contains all the relevant information to perform global routing, and/or enable realistic estimation of design congestion.

We now introduce some notations and definitions that will be used in the remainder of this paper. Typically, global routers overlay a regular grid (g-cells) on the chip image, and construct a global routing graph (Figure 1). Each node in the graph represents a g-cell in the layout, and an edge (g-edge) represents the boundary between adjacent g-cells. To represent/evaluate a global routing solution, we use the following notations, defined for each g-edge (e) in the global routing graph:

- c_e : Total or maximal capacity of edge e
- b_e : The routing blockage on edge e
- w_e : The routing demand or wire occupancy on edge e

2.2 Modeling Local Routing Resource Usage

One of the issues with existing global routing based congestion analyzers, especially the ones using academic global routers, is that they fail to consider the effect of local congestion due to *intra*-g-cell routes. To account for local routing resource usage, the congestion analyzers used for the DAC 2012 contest adopted a simple, yet effective technique based on the pin-density within each g-cell [18].

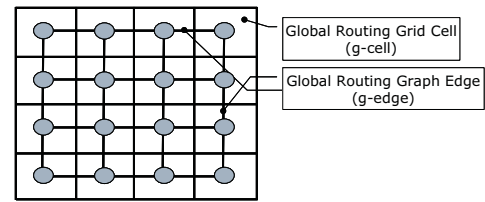


Figure 1: Global routing graph.

Briefly, before performing global routing, the pin-count, say N , within each g-cell is determined. Then, the available capacity of the g-cell (or, associated g-edges) is reduced by $N \times Pin_Blockage_Factor$, where *Pin_Blockage_Factor* is a pre-determined technology specific parameter. Note that the capacity is only reduced on metal layers $M2$ and $M3$ to account for resource usage due to local routes.

2.3 Congestion Metrics

In addition to congestion maps, optimization tools and designers rely on a congestion metric to evaluate the routability of a particular design point. A good congestion metric should be able to accurately score or represent the design congestion as visualized by a congestion map.

2.3.1 Total Overflow and Maximal Overflow

Two commonly used congestion metrics are the total overflow (TOF) and maximal overflow (MOF). Let, $OF(e) = \max(w_e + b_e - c_e, 0)$ be the overflow of a g-edge e . Then, $TOF = \sum OF(e)$ and $MOF = \max(OF(e))$ across all the g-edges in the global routing graph.

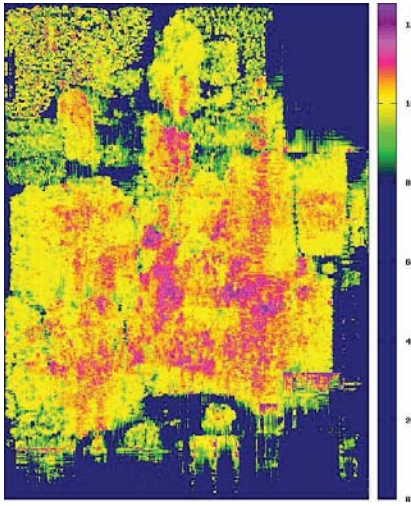
A key issue with these metrics is that they do not provide an accurate view of the design routability. For example, looking at a given TOF value, say, 1283, a designer cannot determine if the g-edges with excess demand are all concentrated in a local region or evenly distributed across the chip image. In the former case, the design might have routability concerns, whereas, in the latter case, the design might still be routable, as the overflow could be easily absorbed by neighboring regions or resolved during subsequent routing phases. In addition, overflow based metrics can fluctuate greatly, based on the number of g-edges, metal layers, etc.

2.3.2 ACE: Average Congestion of g-cell Edges

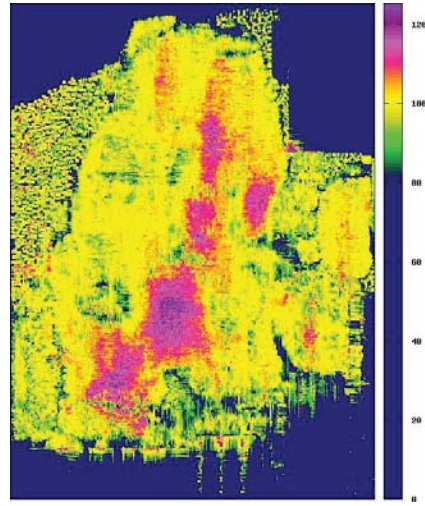
In light of the drawbacks with existing overflow based metrics, the DAC 2012 contest adopted a new metric – the Average Congestion of g-cell Edges (ACE) [18], which is based on the histogram of g-edge congestion. Let, $Cong(e) = 100 \times (w_e + b_e) / c_e$ represent the congestion (in percentage) of a g-cell edge (or, g-edge) e . Then, the new metric, denoted as $ACE(x)$, computes the average congestion of the top $x\%$ congested g-edges. Using different values of x , say, $x \in \{0.5, 1, 2, 5, 10\}$, a set of ACE values can be computed, which together provide an accurate view of the design congestion. For example, $ACE(0.5)$ or $ACE(1)$ provide a highly local view, representing the most congested regions in the design. Whereas, $ACE(5)$ or $ACE(10)$ give a broader view of the design congestion.

2.3.3 Total Overflow versus ACE

In this section, we demonstrate the limitations of the total overflow (TOF) metric in predicting design routability, and



Ripple [6] with total overflow of 542786



SimPLR [10] with total overflow of 514614

Figure 2: Congestion maps and total overflow values for Ripple and SimPLR placement solutions on superblue12. The regions colored purple indicate congestion hot-spots – regions with the greatest contention for wiring resources.

Placer	$ACE(0.5)$	$ACE(1)$	$ACE(2)$	$ACE(5)$	$ACE(10)$
Ripple	126.23	123.00	120.62	114.32	109.10
SimPLR	130.89	126.34	123.17	118.97	113.49

Table 1: The $ACE(x)$ metric for Ripple and SimPLR placement solutions on superblue12, for $x \in \{0.5, 1, 2, 5, 10\}$.

the ability of the ACE metric to accurately represent the design congestion as visualized by a congestion map. For our experiments, we evaluated the placements from two of the top-performing teams (Ripple [6] and SimPLR [10]) in the ISPD-2011 routability-driven placement contest [17].

Recall, TOF was the primary evaluation metric during the ISPD-2011 contest. Let us consider the Ripple and SimPLR placements for the design *superblue12*. The TOF values as reported by the ISPD-2011 contest evaluator are: Ripple (542786) and SimPLR (514614), indicating that the SimPLR placement has better routability.

Figure 2 shows the corresponding congestion maps. From Figure 2, we make two observations: First, on this design Ripple seems to spread more than SimPLR. As a result, the congestion hot-spots for Ripple are more distributed than the hot-spots for SimPLR. Second, the regions colored purple, with congestion $> 120\%$ are much larger in the SimPLR congestion map as compared to Ripple. Based on these two observations and our experience on industry designs, we believe that the SimPLR placement is worse in terms of routability as compared to the Ripple placement. However, this is not captured by the TOF metric, highlighting its limitations.

Next, we apply the ACE metric on the same routing solutions as in Figure 2, and obtain the results shown in Table 1. From Table 1, we see that $ACE(0.5)$ and $ACE(1)$ for SimPLR are higher than that for Ripple, implying that the hot-spots in the SimPLR congestion map have higher congestion than the ones in the Ripple congestion map. More-

over, all the ACE values for SimPLR are worse than Ripple. Together, they indicate that SimPLR has worse routability compared to Ripple on this design. This correlates well with our observations from the congestion maps.

These observations indicate that ACE can represent the congestion map with higher fidelity than the TOF metric, thereby validating the choice for using it as the metric to evaluate the routability of the placement solutions during the DAC 2012 contest.

3. THE DAC 2012 CONTEST METRIC

The DAC 2012 contest used three key metrics to evaluate the quality of the placement solutions:

- The total half-perimeter wire length (HPWL) of the placement.
- The global routing based congestion, as represented by the ACE metric.
- The run-time to generate a placement solution.

3.1 Routability Metric

Let,

- Peak_Weighted_Congestion (PWC):

$$PWC = \frac{\sum(K_x \times ACE(x))}{\sum K_x}, \quad x = 0.5, 1, 2, 5$$

where, K_x is the weight associated with $ACE(x)$.

- Routing_Congestion (RC):

$$RC = \max(100, PWC)$$

Based on the HPWL, and the global routing based congestion (RC), the metric to score/evaluate the routability of the placement solution is given by:

$$Routability_Metric = HPWL \times (1 + PF \times (RC - 100))$$

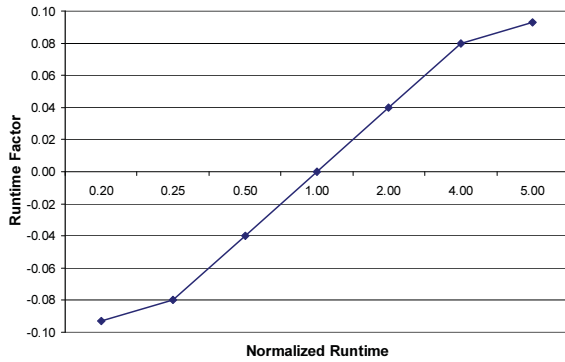


Figure 3: Run-time factor versus normalized run-time. A factor of two speed-up (Normalized Runtime = 0.5) gives a 4% wire length advantage (Run-Time Factor = -0.04).

where, PF is the penalty factor that scales the HPWL to account for routing congestion.

Let, $K_x = 1.0, \forall x$, and the penalty factor, $PF = 0.03$, then the interpretation of the routability metric is that for every 1% excess routing congestion ($> 100\%$), there is a 3% wire length penalty.

3.2 Run-time Factor

To encourage placer efficiency, the placement run-time was also considered in the overall contest metric. This was achieved by using a run-time factor that awards or penalizes the placers based on their run-time. For each design in the benchmark suite, the wall times for all the placers were measured. Let, $Normalized\ Runtime = Placer_WallTime / Median_Placer_WallTime$. Then, the run-time factor for a placer, on a particular design, is given by Figure 3. From Figure 3, a placer has a $\pm 4\%$ wire length advantage for a factor of two speed-up or slow-down. In addition, the maximum run-time factor is set to 10%, limiting the effect of the run-time on the overall contest metric.

3.3 Contest Metric

The overall contest metric, accounting for routing congestion and run-time, is given by the scaled wire length of the placement solution:

$$Contest_Metric = HPWL \times (1 + PF \times (RC - 100)) \times (1 + Runtime_Factor)$$

4. THE DAC 2012 BENCHMARK SUITE

Table 2 presents key statistics of the designs in the DAC 2012 benchmark suite. For each design, the reported statistics are:

- *Total Nodes*: The total number of nodes (movable + fixed).
- *Movable Nodes*: The number of movable nodes.
- *Terminal Nodes*: The number of fixed “terminal” nodes. There can be no overlap between the movable and terminal nodes.
- *TerminalNI Nodes*: The number of fixed “terminalNI” nodes. Overlap is allowed between the movable and terminalNI nodes.

- *Total Nets*: The total number of nets.
- *Total Pins*: The total number of pins.
- *Design Util.*: The design utilization, defined as the ratio of the total area of the movable and terminal nodes to the area of the placement region.
- *Design Den.*: The design density, defined as the ratio of the total area of the movable nodes to the available free-space in the design. Where, free-space is given by the area of the placement region, less the total area of the terminal nodes.

All the designs in the benchmark suite are derived from industrial designs, and converted from the IBM internal data format to the GSRC Bookshelf placement format [3,4]. The height of all the movable and terminal nodes are an integer multiple of the standard-cell circuit row height. This allows every node to be placed aligned to the circuit rows in the design. In addition, all the macro-blocks (nodes with height $>$ circuit row height) are fixed in-place. Hence, the only movable nodes in all the designs are the standard-cells. Movable macro-blocks greatly complicate the placement problem. In addition, they can significantly alter the congestion profile. To start with, we would like to evaluate the performance and routability of the placers when they are allowed to move only the standard-cells.

In order to preserve the integrity of the industrial designs, and enable realistic estimation of design congestion (e.g., global routing based congestion analysis), the Bookshelf placement format was augmented with some new features. These features were introduced during the ISPD-2011 placement contest [17], and are described below.

4.1 Non-rectangular Fixed Nodes

A subset of the pre-placed IP blocks in the designs are not rectangular. One such node is shown in Figure 4, where the light-red shaded (solid) box with blue outline represents a rectangular fixed node, whereas, the large gray object (hatched lines) represents a non-rectangular fixed node. To represent the non-rectangular nodes in the Bookshelf format, each non-rectangular node is fragmented into a set of single-circuit row high rectangular component shapes. This is depicted by the narrow boxes within the overall gray outline in Figure 4. A new benchmark file *circuit.shapes*, then specifies the component shape definitions for each non-rectangular node. In addition, the other files in the benchmark are modified to account for pin-offsets, node dimensions, node coordinates, etc.

4.2 RLM Pins and terminal_NI nodes

In a hierarchical design methodology, there are multiple lower level random logic macros (RLMs) that are integrated at the top level to yield the final design. To enable interconnection across different levels of the hierarchy, designs use special pins, which we denote as RLM pins. RLM pins typically reside on the upper-level metal layers associated with a particular macro. To represent such pins in the Bookshelf format, a new node class named “terminal_NI” is introduced. Each terminal_NI node contains one RLM pin. For placement, the standard-cells can be placed “below” the terminal_NI nodes without violating any legality constraints. During global routing, the pins associated with the terminal_NI nodes reside on an upper-level metal layer (e.g., M5),

Design	Total Nodes	Movable Nodes	Terminal Nodes	Terminal_NI Nodes	Total Nets	Total Pins	Design Util. (%)	Design Den. (%)
superblue19	522775	506097	286	16392	511685	1714351	78	49
superblue14	634555	567840	44743	21972	619815	2049691	72	50
superblue16	698741	680450	419	17872	697458	2280931	69	46
superblue9	846678	789064	37574	20040	833808	2898853	73	47
superblue3	919911	833370	55033	31508	898001	3110509	73	42
superblue11	954686	859771	67303	27612	935731	3071940	79	40
superblue6	1014209	919093	65316	29800	1006629	3401199	73	43
superblue2	1014029	921273	59312	33444	990899	3228345	76	28
superblue12	1293433	1278084	8953	6396	1293436	4774069	56	44
superblue7	1364958	1271887	66995	26076	1340418	4935083	76	58

Table 2: Design statistics of the DAC 2012 benchmark suite.



Figure 4: Non-rectangular versus rectangular fixed nodes. A light-red shaded box with blue outline represents a rectangular fixed node, whereas, the large gray object represents a non-rectangular fixed node. Note: The non-rectangular fixed node is represented as a set of rectangular component shapes (narrow boxes within the overall gray outline).

and should be connected accordingly. Terminal_NI nodes and associated RLM pins are depicted in Figure 5.

4.3 Routing Information

To enable contestants to perform global routing, and/or other means of congestion analysis, a new file *circuit.route* is created for each design in the benchmark suite. This file has the following information:

- Global routing grid (grid size, g-cell dimensions, etc.)
- Number of routing layers.
- Maximal g-edge capacity for each routing layer.
- Wire width and spacing for each routing layer.
- Via specifications.
- Routing blockage information.
- Routing layer for RLM pins.

Figure 6 shows a sample *circuit.route* file reflecting an industrial wiring stack with nine metal layers. Layers M1–M4 have 1× width and spacing, layers M5–M7 have 2× width and spacing, and layers M8–M9 have 4× width and spacing.

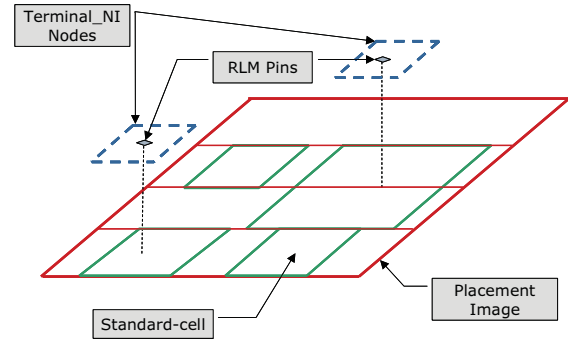


Figure 5: Terminal_NI nodes and associated RLM pins. Overlap is allowed between the movable and terminal_NI nodes during placement. The RLM pins on terminal_NI nodes reside on an upper-level metal layer (e.g., M5) during global routing.

route 1.0									
Grid	:	304	403	9					
VerticalCapacity	:	0	80	0	80	0	80	0	80
HorizontalCapacity	:	0	0	80	0	80	0	80	0
MinWireWidth	:	1	1	1	1	2	2	2	4
MinWireSpacing	:	1	1	1	1	2	2	2	4
ViaSpacing	:	0	0	0	0	0	0	0	0
GridOrigin	:	18	18						
TileSize	:	40	40						
BlockagePorosity	:	0							
Header Section									
NumNITerminals	:	2							Terminal_NI Section
p0	5								# All the pins associated with nodes p0/p1 are on metal layer M5 for routing
p1	5								
NumBlockageNodes	:	2							Blockage Section
o44	4	1	2	3	4				# o44/o2407 block 4 metal layers within the overlapping g-cells. These are M1-M4.
o2407	4	1	2	3	4				

Figure 6: A sample *circuit.route* file with information to setup and perform global routing for a given placement. This example shows a layer stack with nine metal layers, along with varying width and spacing across layers.

4.4 Floorplan Layouts

Figures 7 through 9 depict the floorplans for all the designs in the benchmark suite, plotting only the fixed (“terminal”) nodes. The light-red shaded boxes with blue outline represent the rectangular fixed nodes and the dark-gray shaded boxes represent the non-rectangular fixed nodes in the design. Figures 7 through 9 and Table 2 demonstrate the varying characteristics of modern ASIC designs. For example, all the fixed nodes in *superblue12* are pushed to the periphery of the placement region. As a result, the placer has a large amount of contiguous free-space in and around the center of the placement region. On the other hand, the fixed nodes in *superblue9* and *superblue7* fragment the placement region into multiple sub-regions. The design utilization of *superblue2* is greater than 75%, on account of the large fixed nodes, whereas, its design density is 28%, indicating that there is also abundant free-space to place the movable nodes. All these features offer different challenges to routability-driven placement algorithms. In addition, two of the designs *superblue12* and *superblue7* have over a million movable nodes, and are good benchmarks to test the scalability of the placers.

5. CONCLUSIONS

Routing congestion is a key issue in nanometer-scale physical synthesis flows. Numerous factors contribute to routing congestion in modern designs, e.g., increased use of embedded IPs, complicated logic structures, a high-performance layer stack that is non-uniform in nature, per-layer routing blockages, etc. As a result, routing congestion needs to be accurately modeled and addressed across the entire physical synthesis tool stack. In this regard, routability-driven placement is a highly relevant and challenging problem.

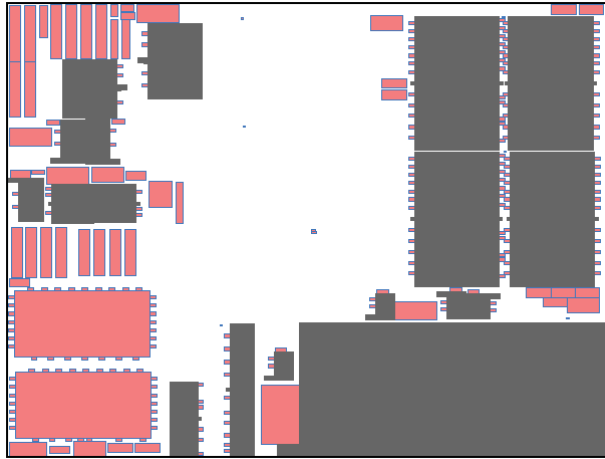
This paper describes the DAC 2012 routability-driven placement contest and benchmark suite. All designs in the DAC 2012 benchmark suite are derived from modern industrial ASICs, and contain information to perform both placement and routing. In addition, the contest provides standard metrics, as well as an accurate congestion analysis framework to evaluate and compare the routability of placement algorithms. We hope that a set of challenging benchmarks, along with a standard, publicly available evaluation framework will further advance research in placement and routing for nanometer-scale designs.

Acknowledgments

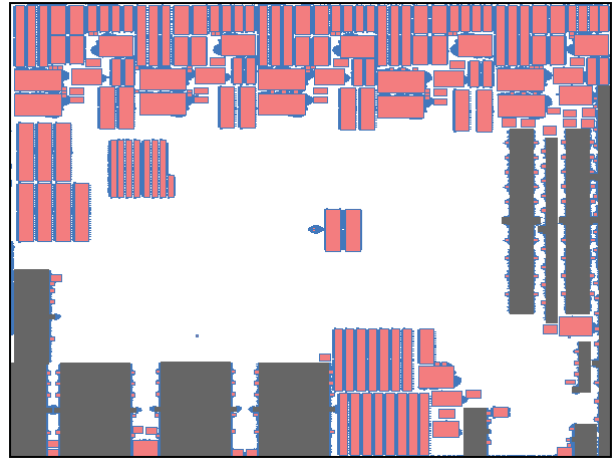
A key requirement for the contest was the availability of fast, high-quality global routers to evaluate the placement solutions. The authors would like to thank Wen-Hao Liu and Prof. Yih-Lang Li, Department of Computer Science, National Chiao Tung University, Taiwan; Jin Hu and Prof. Igor Markov, Department of Electrical Engineering and Computer Science, University of Michigan, Ann Arbor; for their time and effort to qualify their respective routers, NCTUgr and BFG-R, to be used for the DAC 2012 contest. Thanks also go to our colleagues at IBM, Bertram Bradley, Randy Darden, Gi-Joon Nam and Shyam Ramji for all the discussions and their help in releasing the benchmarks.

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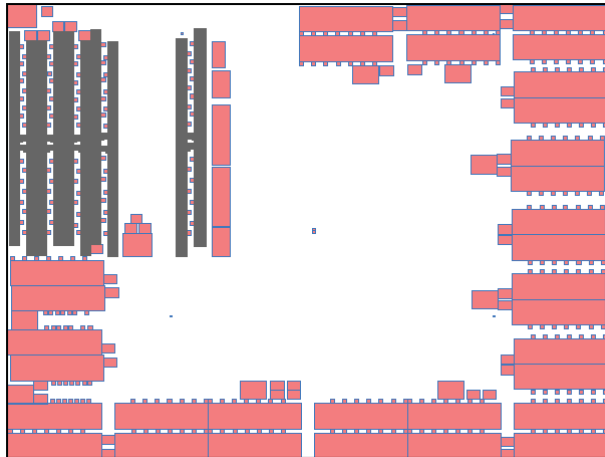
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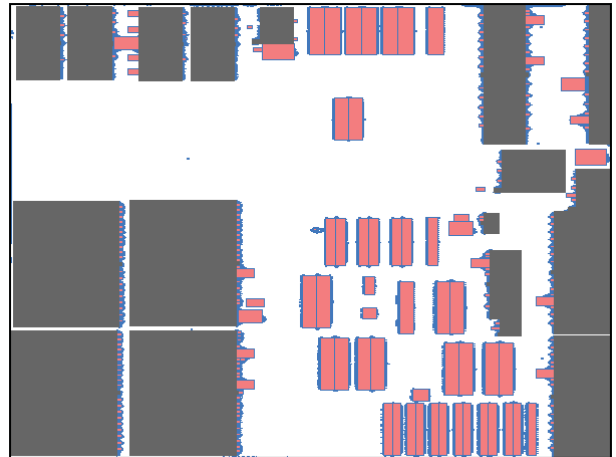
superblue19



superblue14

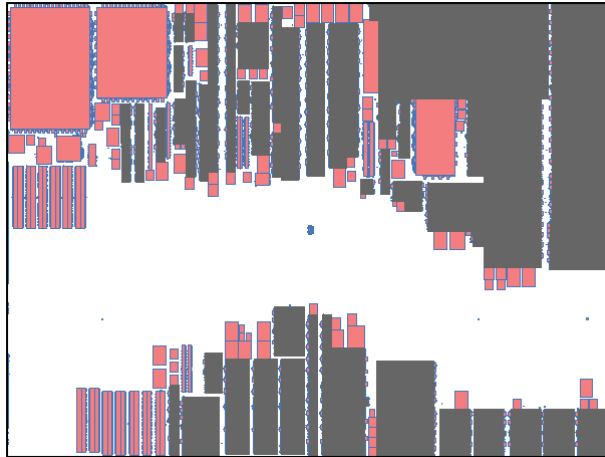


superblue16

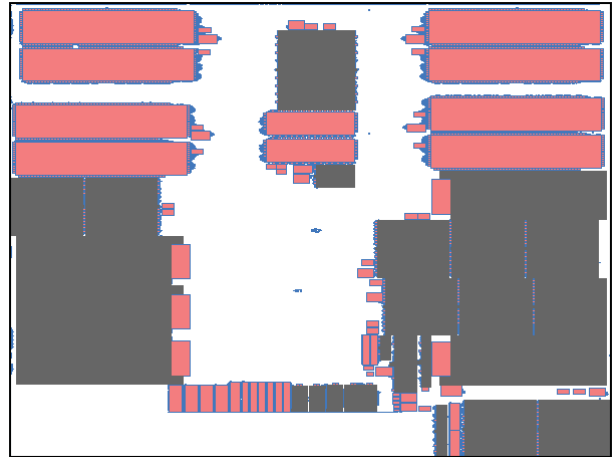


superblue9

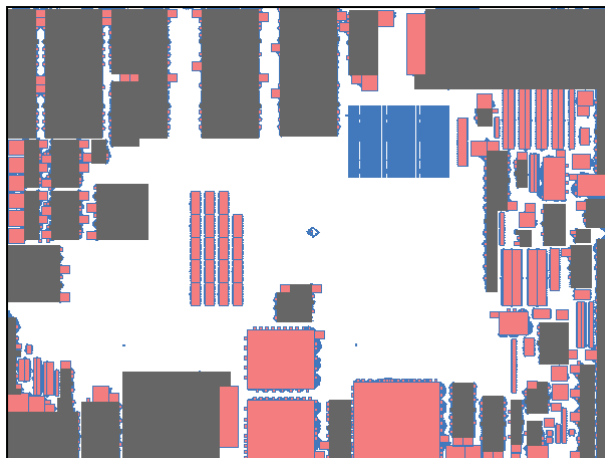
Figure 7: Floorplan layout figures of designs superblue19, superblue14, superblue16 and superblue9. The light-red shaded boxes with blue outline represent the rectangular fixed nodes and the dark-gray shaded boxes represent the non-rectangular fixed nodes in the design.



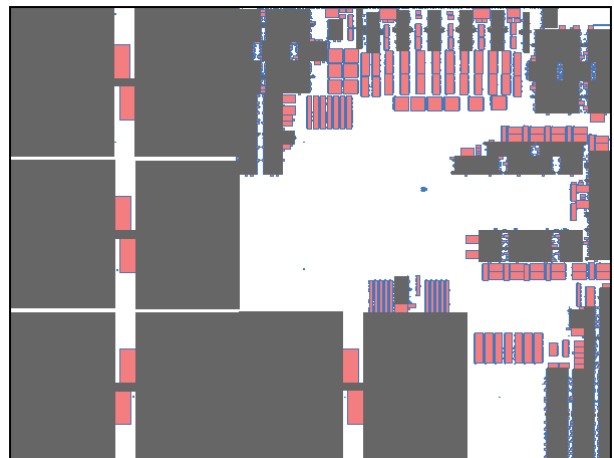
superblue3



superblue11

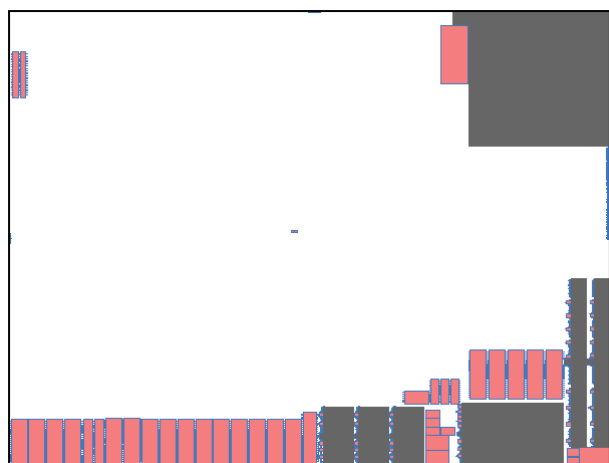


superblue6

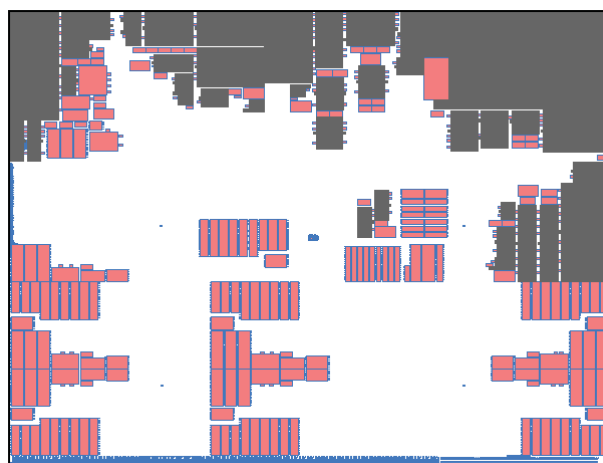


superblue2

Figure 8: Floorplan layout figures of designs superblue3, superblue11, superblue6 and superblue2. The light-red shaded boxes with blue outline represent the rectangular fixed nodes and the dark-gray shaded boxes represent the non-rectangular fixed nodes in the design.



superblue12



superblue7

Figure 9: Floorplan layout figures of designs superblue12 and superblue7. The light-red shaded boxes with blue outline represent the rectangular fixed nodes and the dark-gray shaded boxes represent the non-rectangular fixed nodes in the design.